UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing – Fall 2026

Homework 6

Profs. Kamalabadi, Snyder **Due: Friday, Oct 9, 11:59pm on Gradescope**

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1. Determine the Fourier transform of the following functions:
   (a) $\delta(5t-2)$
   (b) $\cos(\Omega_0 t+\phi_0)$, where Ω_0 and ϕ_0 are known real numbers.
   (c) $u(t)-u(t-T)$, where T is a constant.

2. Let $x[n]$ be an arbitrary sequence, not necessarily real-valued, with DTFT $X_d(\omega)$. Express the DTFT of the following sequences in terms of $X_d(\omega)$
   a) $x^*[n]$
   b) $x^*[-n]$

3. Find $x[n]$ for each DTFT given below:
   (a) $X_d(\omega)=e^{-j\omega 3}+2e^{-j\omega 10}$
   (b) $X_d(\omega)=e^{-j\omega 3.5}$
   (c) $X_d(\omega)=\cos^2(\omega)$

4. Compare the DTFT and z-transform of the following sequences:
   (a) $x[n]=3^{-n}u[n]$
   (b) $x[n]=e^{jn\frac{\pi}{4}}u[n]$

5. A causal LSI system is described by the difference equation: $y[n]-y[n-1]=x[n]$.
   (a) Determine the system's transfer function $H(z)$
   (b) Determine the system's unit pulse response $h[n]$
   (c) Determine the system's frequency response $H_d(\omega)$; is $H_d(\omega)=H(z)|_{z=e^{j\omega}}$? if not, explain why.

6. An LSI system is described by the difference equation

$$y[n]=x[n]+x[n-10]$$

   (a) Compute and sketch its magnitude and phase response
   (b) Determine its output to inputs
      i. $x[n]=\cos\frac{\pi}{10}n+3\sin\left(\frac{\pi}{3}n+\frac{\pi}{10}\right)$
      ii. $x[n]=10+5\cos\left(\frac{2\pi}{5}n+\frac{\pi}{2}\right)$

7. The frequency response of an LSI system is

$$H_d(\omega) = \omega e^{j \sin \omega}, \qquad |\omega| \leq \pi \ .$$

Determine the system output $y[n]$ for the following inputs:

(a) $x[n] = 3 - 10e^{j(\frac{\pi}{4}n + 45^\circ)} + j^n$

(b) $x[n] = 3 + 10\cos(\frac{\pi}{4}n + 45^\circ) - j^n$.

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